



Upper Extremity Orthopedic Treatment

FT300 PT500

Wrist & Hand

- Common conditions

Learning Objectives

- Describe all special tests (indication, procedure, positive finding, explanation of results) in special tests document
- Perform all special tests in special tests document using full NEER protocol
- Describe definition, MOI, affected tissue, signs and symptoms for UCL of MCP 1 sprain; FPL tenosynovitis; CMC 1 Osteoarthritis; Finger arthritis (OA & RA); Dupuytren contracture; Trigger finger; De Quervain's tenosynovitis; TFCC injury; Fracture
- Describe treatment plan (history, physical assessments, clinical impression and approach to treatment, precautions, and home care) for UCL of MCP 1 sprain; FPL tenosynovitis; CMC 1 Osteoarthritis; Finger arthritis (OA & RA); Dupuytren contracture; Trigger finger; De Quervain's tenosynovitis; TFCC injury; Fracture



Special Tests

Special tests



A reminder that all special tests for the wrist & hand are in the Orthopedic Treatment Special Tests notes document.

Refer to this handout.





Common conditions

Common conditions

UCL of MCP 1 Sprain

Flexor Pollicis Longus Tenosynovitis

First CMC osteoarthritis

Finger arthritis (RA & OA)

Dupuytren contracture

Trigger finger

De Quervain's tenosynovitis

TFCC injury

Fracture

UCL of MCP 1 Sprain (aka gamekeeper's thumb, skier's thumb)

UCL MCP 1: About

Most common cause of ulnar sided thumb pain

Injured by valgus stress to MCP 1

May be due to:

- Acute trauma
- Repetitive stress
- Secondary injury (e.g. from CMC 1 OA)

Game Keeper's Thumb



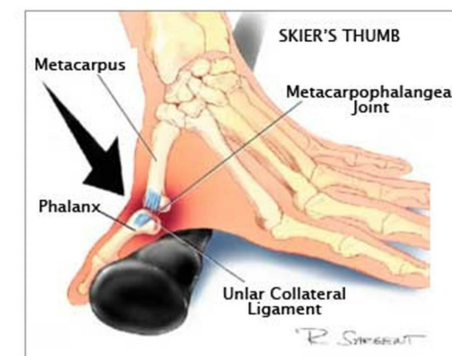
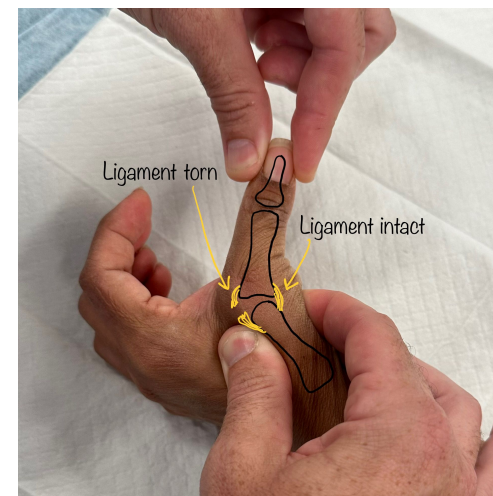
UCL MCP 1: History

If due to acute trauma (a fall) or repetitive stress (occupational demand), the person can usually identify the cause

If secondary (from CMC 1 OA), the cause will be less clear or unknown

Symptoms:

- Ulnar-sided thumb pain
- Difficulty gripping and pinching



MECHANISM OF INJURY RESULTING IN A SKIER'S THUMB

UCL MCP 1: History



Key questions:

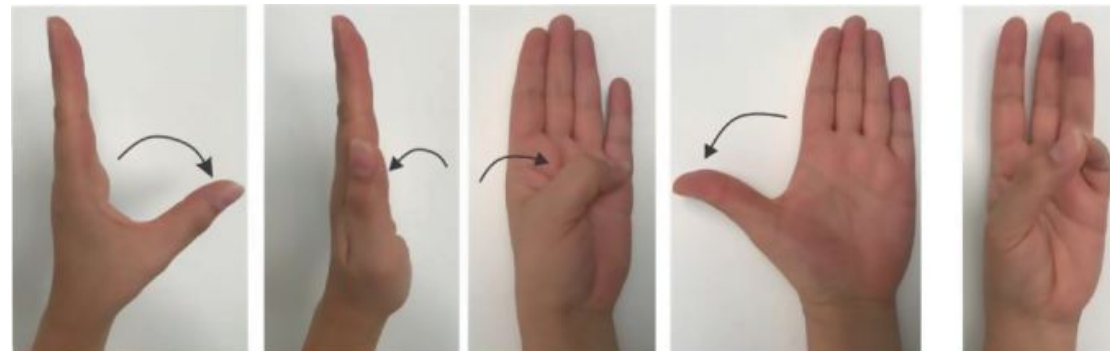
- If suspecting arthritis, it is important to be clear if it is OA or RA that is present

UCL MCP 1: Physical assessments



ROM:

- Wrist ROM is probably fairly normal but may show signs of muscle tightness in reduced ranges
- Thumb ROM may show reduced extension due to thenar contracture; the end range of abduction is likely to be painful.
- Thumb RROM in opposition is likely to be painful and weak



Abduction-adduction

Flexion-extension

Opposition

UCL MCP 1: Physical assessments



Muscle testing:

- Muscle shortening may occur in thenar muscles (e.g. flexors and adductor)
- Muscle weakness may occur due to pain with gripping or pinching motions (Flexor pollicis longus and brevis; abductor pollicis longus and brevis; opponens pollicis)

UCL MCP 1: Physical assessments



Special tests:

- UCL stress test of the thumb is positive. Pain without laxity is a mild injury; laxity with a firm end feel is a moderate injury. Laxity with a soft end feel is a severe injury.



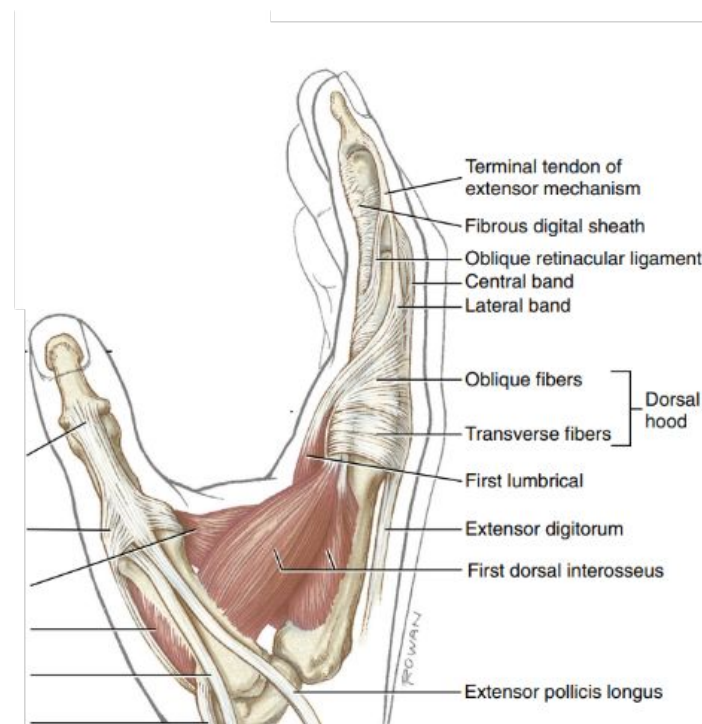
UCL MCP 1: Management

Severe injuries (rupture)

- Usually require surgery

Mild or moderate injuries

- Brace, splint, orthosis for 2-6 weeks
- Gentle AROM encouraged during protection
- Pain-free muscle setting and isometrics
- Focus on first dorsal interosseous muscle



UCL MCP 1: Massage treatment

Overview of treatment goals for ligament sprain rehabilitation

1. Promote a functional scar and reduce adhesions at site of ligament injury
2. Restore neuromuscular control and resting muscle tone of the muscles around the joint
3. Restore normal range of motion (muscle balance and joint integrity around a joint)
4. Treat compensatory structures to restore overall function

Cautions

Do not perform valgus or medial gapping of 1st MCP
Be cautious with passive or end range thumb abduction and circumduction

UCL MCP 1: Expected outcomes

Non-surgical injuries (mild or moderate), if protected adequately and provoking factors are eliminated or significantly reduced, may recover in 3-10 weeks.





FPL Tenosynovitis

FPL: About

What is tenosynovitis?

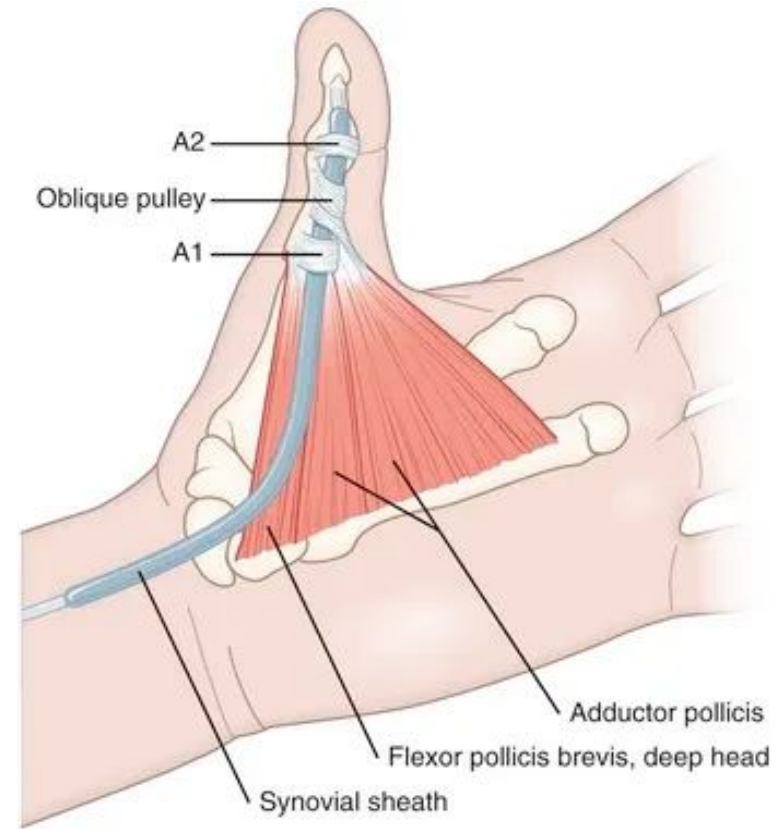
FPL sheath (aka radial bursa) runs from proximal to carpal tunnel to DIP

Overuse condition

- Gradual onset

Main complaint

- Pain
- Most commonly at A1 pulley (see image)



FPL: History

Palmar thumb pain, most commonly following activity requiring thumb movement

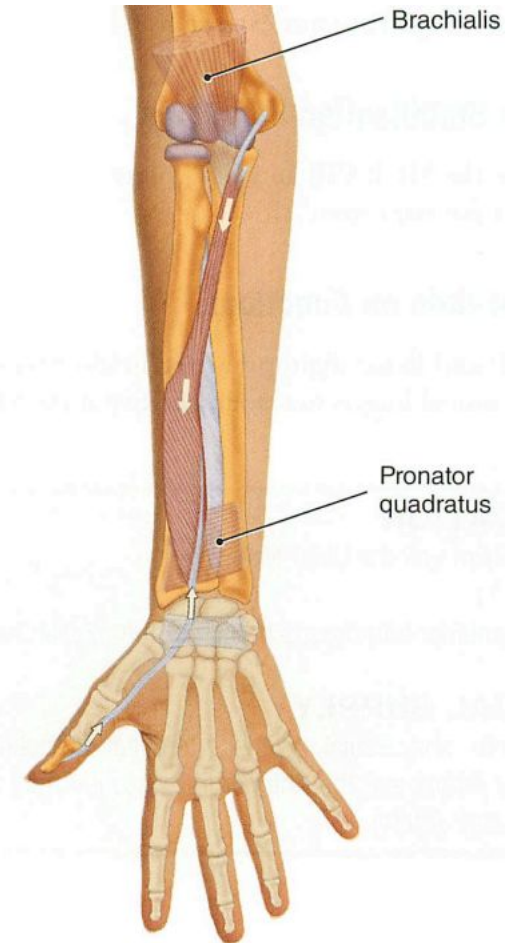
AROM generally worse than RROM...why?

Common activities:

- Gripping
- Rock climbing
- Texting

Key questions:

- New activity or recent increase?



FPL: Physical assessments

ROM:

- AROM of thumb movement is painful, especially flexion and extension. A clicking may be present with thumb flexion or extension.
- Isometric testing - e.g. RROM - is often pain-free.

FPL: Physical assessments

Muscle testing:

- Muscle shortening is not common
- Muscle weakness is not common. Testing the strength of FPL or FPB may be useful to rule out strain to these muscles.

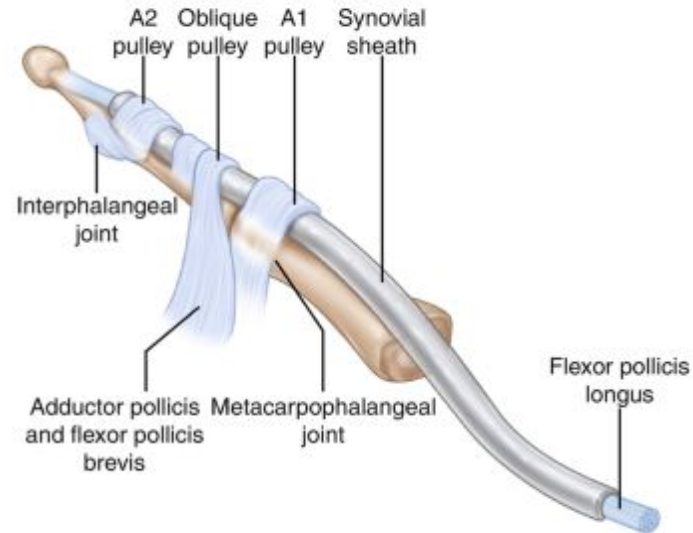
FPL: Physical assessments

Special tests:

- UCL & RCL of the 1st MCP may be used to rule out ligament sprain

FPL: Management

Activity modification
Pain modulation
Exercises



FPL: Management

Activity modification

- Reduce or temporarily remove provoking factor
- Taping or splinting

Pain modulation

- NSAIDs, ice

Exercises

- Isometrics are safe to perform
- Avoid exercises that require movement of the thumb
 - Person ready for AROM or concentric exercises once able to perform 10 reps of 60 seconds isometric holds without symptoms

FPL: Massage treatment

Overview of treatment goals for tenosynovitis rehabilitation

1. Reduce adhesions around the tendon sheath
2. Promote normal resting muscle tone of the affected muscle
3. Promote normal joint play and reduced fascial restrictions in area
4. Treat compensatory structures to restore overall function

Cautions

Do not friction or put direct pressure down into the FPL tendon sheath
Avoid active or passive movement of the thumb unless clearly pain-free
Avoid heat directly over the tendon sheath which may be inflamed



FPL: Expected outcomes

Expected timelines are highly variable. It is very difficult to offload the thumb as it is used in so many vital ADLs.

Adherence to modifications and protection of the area are most important in determining the timeline for recovery.

Mild cases in adherent individuals may be cleared up in 10-21 days in the best case scenarios.



CMC 1 OA

CMC: About

Common in older adults

CMC 1 OA can be quite disabling due to the importance of the thumb in most functional activities

(The thumb is 40-50% of hand function...the hand is 90% upper limb function)



CMC: About

Pain is normally felt around the anatomical snuffbox and/or thenar eminence, especially with thumb movement.



CMC: History

Worse in morning, but improves with some movement and use of heat

Pain with activities that compress the CMC 1:

- Gripping
- Pinching
- Radial deviation

Pain leads to disuse and then loss of function

While swelling may occur, acutely swollen, red, taut, warm joints are not expected (they suggest RA instead of OA)

It's important to ask about bony changes / bone spurs in the joint

CMC: Physical assessments

ROM:

- Wrist ROM may have pain with radial deviation, which is a compressive force on the CMC 1. CMC 1 OA is the most likely condition present if thumb pain is recreated during radial deviation of the wrist.
- Thumb ROM may be reduced and painful in all ranges, especially abduction and extension.
- Thumb RROM is painful over the CMC in multiple ranges

CMC: Physical assessments

Muscle testing:

- Muscle shortening may occur in thenar muscles - particularly flexors and adductors
- Muscle weakness may occur in intrinsic hand muscles - thenar group and central compartment group. Pain and disuse may lead to atrophy of the first dorsal interosseous manus muscle.

CMC: Physical assessments

Special tests:

- Joint play assessments may be decreased in the CMC
- UCL & RCL stress test of the thumb may show laxity secondary to CMC degenerative changes

CMC: Management

Managed by:

- Joint protection / education
- Strengthening
- ROM & stretching
- Joint mobilisation
- Orthosis (e.g. image, right)



CMC: Management

Joint protection / education

- Figure out what the provoking activities are and modify
- Modifications may need to be indefinite

Strengthening:

- Isometrics for 1st dorsal interosseous



CMC: Management

ROM & stretching

- Massage to thenar muscles (including adductor pollicis) will help prevent contractures

Joint mobilisation

- As long as there are no bony changes

Orthosis:

- Widely available
- Should be fitted



CMC: Massage treatment

Overview of treatment goals for osteoarthritis management

1. Promote joint nutrition
2. Improve joint function (normal joint play)
3. Improve muscle function to support joint structures
4. Treat compensatory structures to restore overall function

Cautions

Do not mobilise CMC or take thumb to full ROM if osteophytes are present or abnormal bone to bone end feel occurs
Avoid joint compression and be careful with POP

CMC: Expected outcomes

Ultimately, OA is a chronic, progressive condition.

However, improvements in pain and hand function may be achieved in 4-12 weeks.

Weekly treatment for 6 weeks should be an initial treatment plan, with treatment thereafter more spaced out, as long as the person is able to follow a home care plan of pain-free movement and intrinsic hand muscle strengthening.

Review:

Turn to your partner.

One person describe UCL of MCP 1 sprain- what are some common history and physical assessment findings, how do you approach treatment

One person describe CMC 1 OA - what are some common history and physical assessment findings, how do you approach treatment



Finger Arthritis (OA / RA)

OA/RA: About



Finger arthritis may be due to OA or RA

What are some key differences between the two?

What type of nodes are seen in each?



OA/RA: History

Onset is gradual

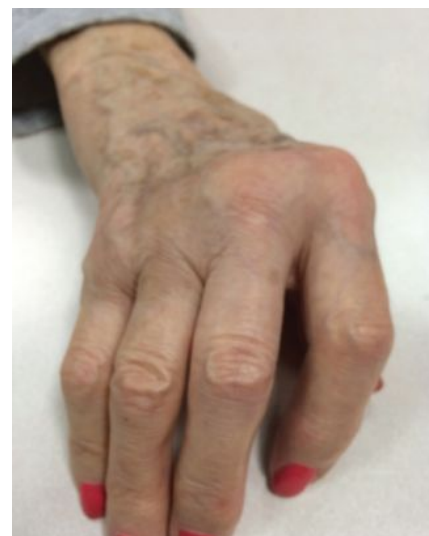
Treatment sought once ADLs are difficult

OA

- morning stiffness under 1 hr
- mainly affects PIP & DIP

RA

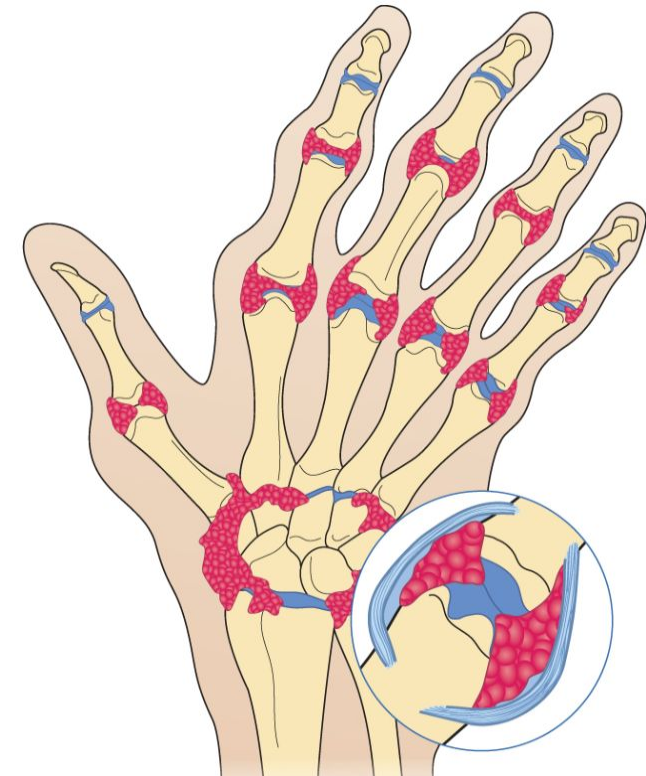
- morning stiffness over 2 hr
- mainly affects MCP



OA/RA: Physical assessments

ROM:

- ROM is typically reduced in multiple ranges in AROM, PROM and RROM



OA/RA: Physical assessments

Muscle testing:

- Muscle shortening may occur in forearm muscles (flexors and extensors) as well as hand muscles
- Muscle weakness may occur in forearm muscles (flexors and extensors) as well as hand muscles

Special tests:

- None

OA/RA: Management

RA:

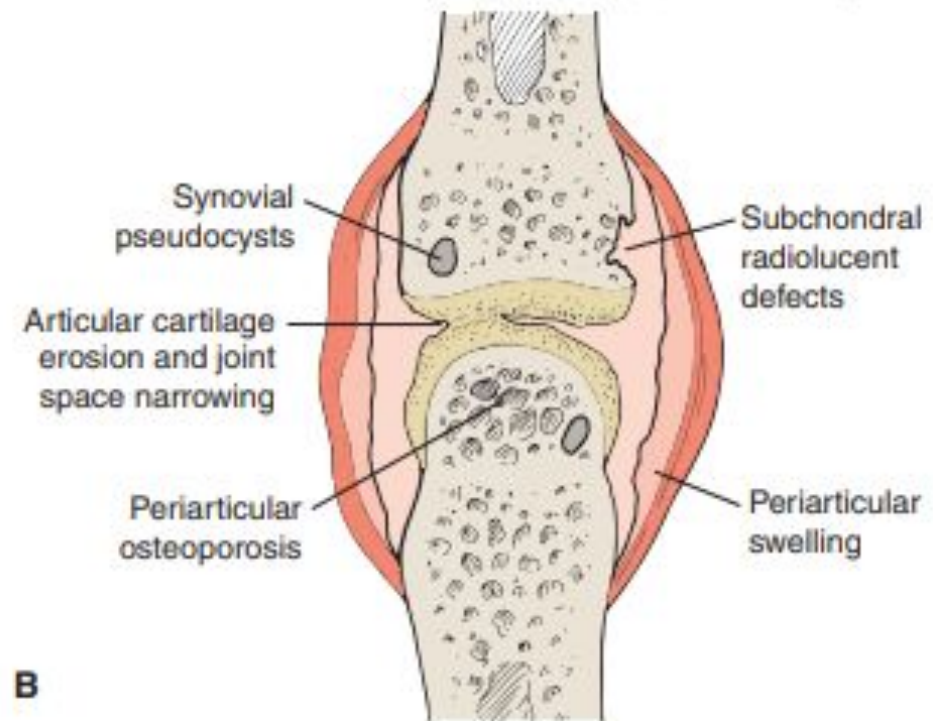
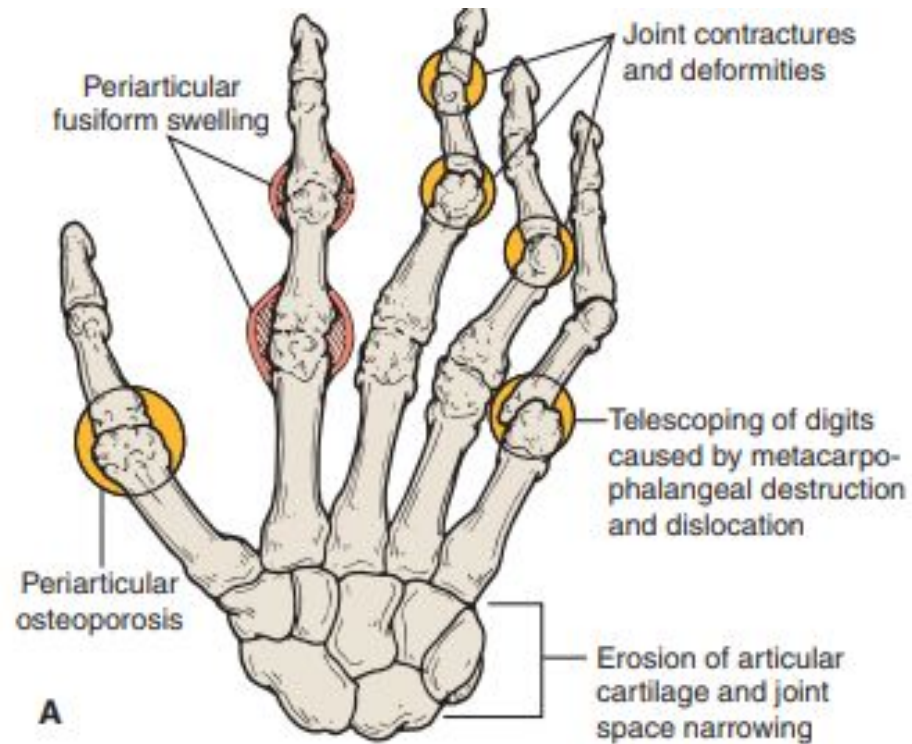
- Joint protection
- Occupational therapist
- Strengthening - intrinsic hand muscles only

OA:

- Heat
- Joint mobilisation (no bone spurs)
- Strengthening



OA/RA: Management



OA/RA: Expected outcomes

Prevention of joint deformities is key for patients with RA, as once irreparable joint damage occurs, it is difficult to manage symptoms or improve function without orthosis or adaptive equipment.

In addition, because patients often do not seek medical care until a deformity has already occurred, abnormal motor patterns and adaptive disuse atrophy are often quite profound, requiring more extensive rehabilitation and reducing the potential to achieve optimal results.



Dupuytren Contracture

Dupuytren: About

A benign, insidious overproduction of fibrous tissue within the palmar fascia

Tissue knots form under the skin and create a thick cord that can bend the fingers

Primarily affects older people

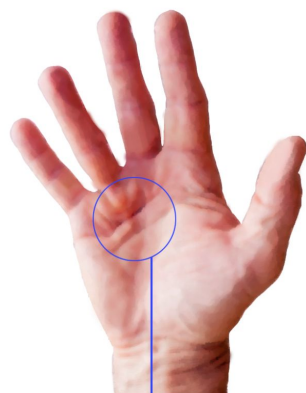
- More common with diabetes mellitus or long term alcohol consumption



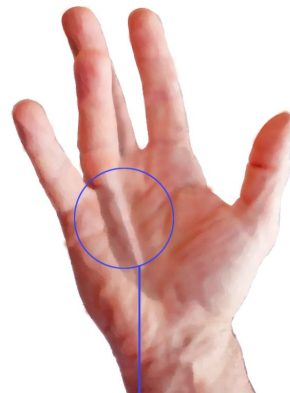
Dupuytren: About

Common impairments:

- Difficulty extending the fingers
- Difficulty using a keyboard
- Inability to flatten the palm



**Nodules and Pitting
May Appear in the Hand**



**Cord Forms
in the Palm**



**Fingers Bend
Toward the Palm**

Dupuytren: History

Onset gradual and progressive (generally slow progressive)

Pain is not expected

Most commonly affects the fascia in line with digits 3-5

Dupuytren: Physical assessments



ROM:

- AROM shows decreased MCP extension, most commonly digits 3 and 4. If severe, PIP extension is also decreased
- PROM be the same as AROM, though slightly more motion is expected
- RROM is likely to show weakness in finger movements, especially extension, due to disuse

Dupuytren: Physical assessments

Muscle testing:

- Intrinsic hand muscles and finger extensors are expected to be weak and/or show poor neuromuscular control due to disuse

Special tests:

- None

Dupuytren: Management

Dupuytren contracture is a long-term, progressive condition that is highly resistant to treatment or change. Even after successful treatment (e.g. surgery), there is a 50% 5 year recurrence rate

Conservative

- Long, slow, low intensity stretches
- Static progressive orthosis

Surgery:

- Fasciectomy, fasciotomy, or injection therapies

Dupuytren: Management



Dupuytren: Massage treatment

Overview of treatment goals for Dupuytren contracture management

1. Promote joint nutrition
2. Improve joint function (normal joint play)
3. Improve muscle function and prevent contractures
4. Treat compensatory structures to restore overall function

Cautions

Do not force the joints beyond their range in extension (long, gentle stretching is more effective than deep stretching or aggressive tissue mobilisation)



Dupuytren: Expected outcomes

Dupuytren's is resistant to change. Short-term improvements (minutes to hours) in mobility may be experienced, but resolution of the complaint will not occur.

It is a progressive, long-term condition. Frequent treatments to help promote and preserve available function are recommended.

Trigger Finger (aka stenosing tenosynovitis)

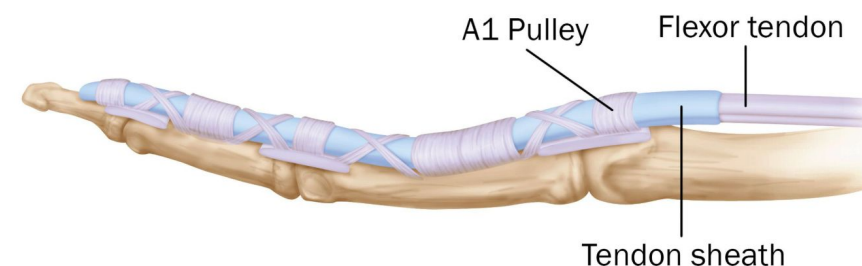
Trigger: About

Caused by friction and inflammation of the FDS with development of an inflammatory nodule near the distal palmar crease

Most commonly affects the A1 pulley (at MCP joint)

When the finger is flexed, the nodule slides proximally under the pulley

When the finger is extended, it must slide back distally...this can be difficult

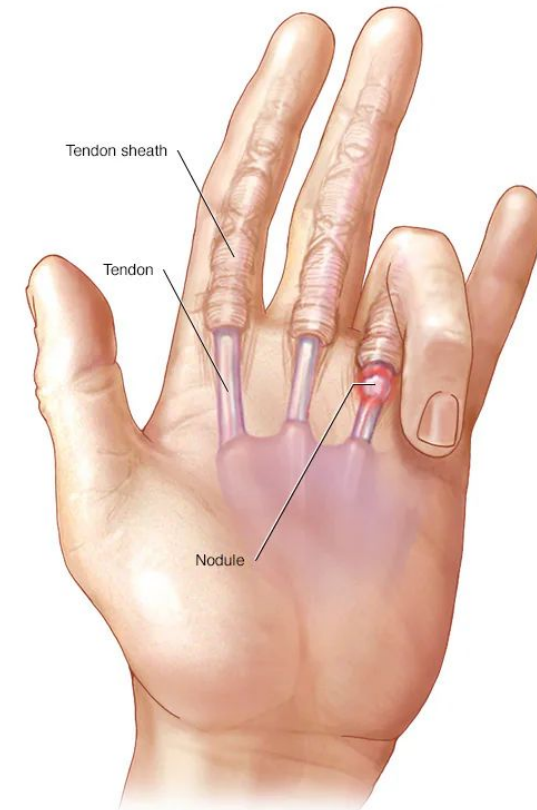


Trigger: About

Progression:

- palpable / audible click or clunk
- pain
- cannot actively extend finger...must use PROM to extend finger
- may not be able to extend finger even passively

Let's watch an example ([link](#))



Trigger: History

Insidious onset (clicking or locking of finger)

Recent history may include activity requiring increased gripping

Small palpable nodule just proximal to the MCP

Tenderness at rest, pain during extension



Trigger: Physical assessments

ROM:

- AROM would show reduced extension with a click, lock, or clunk. This may be observed and/or heard
- PROM would be similar to AROM, with potential difficulty to extend the finger in question
- RROM would likely show reduced finger extension strength (from disuse)

Trigger: Physical assessments

Muscle testing:

- Muscle weakness may occur in the finger extensors from disuse. The muscle belly of FDS is likely to be normal length, despite the tendon gliding problems.

Special tests:

- None

Trigger: Physical assessments

Differentiate from Dupuytren

- Dupuytren's is normally painless, consistent stiffness of the fingers, with dimpling of the palm.
- Trigger finger is normally painful with a clear clunky, clicking, or locking motion of the finger during extension.

Trigger: Management



Reduce inflammation and friction

Activity modification

Splints or orthoses

Corticosteroid injection and orthosis may be given



Trigger: Massage treatment

Overview of treatment goals for trigger finger rehabilitation

1. Promote muscle balance in forearm and hand
2. Improve joint function (normal joint play)
3. Treat compensatory structures to restore overall function

Cautions

Do not work on site or provide heat to the affected nodule (respect acute stage of healing precautions)
Be careful with finger flexion and extension; avoid deep flexion of the finger.

Trigger: Expected outcomes

The vast majority of cases resolve without the need for surgery. Resolution can be difficult - it is important to avoid aggravating the tendon or increasing inflammation or friction. Timelines are variable.

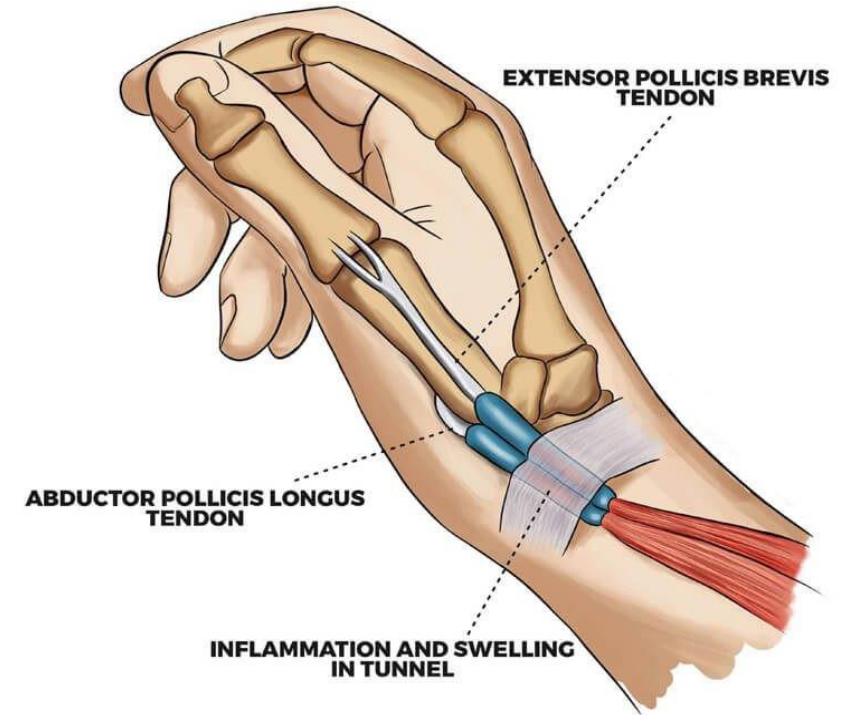
De Quervain's tenosynovitis

DQT: About

Overuse condition, particularly ulnar dev

APL & EPB tenosynovitis

- movement generally worse than isometric



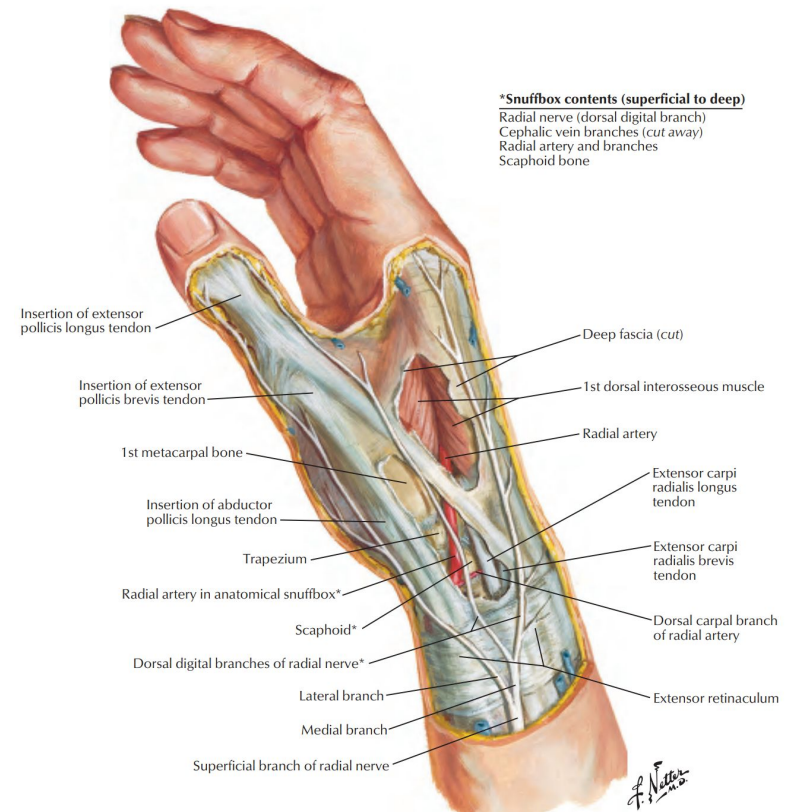
DQT: About

Radial wrist pain from overuse is normally one of 3 conditions:

- 1 / De Quervain's
- 2 / Intersection syndrome
- 3 / Wartenberg syndrome



Lateral (radial) view



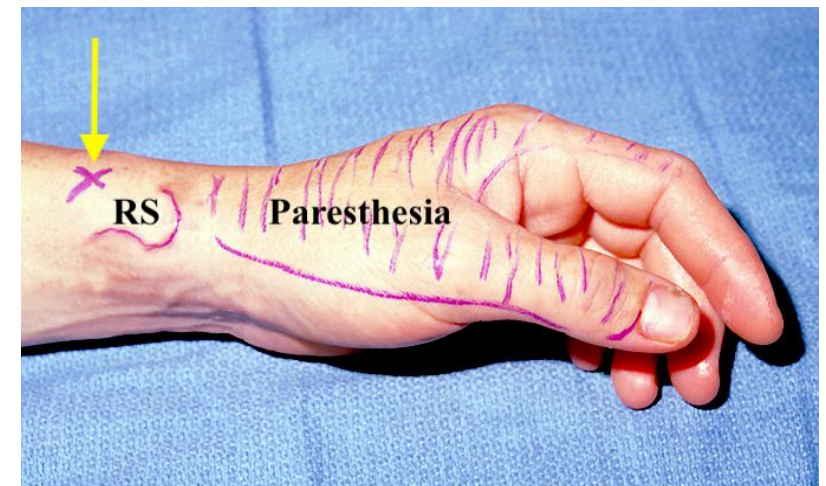
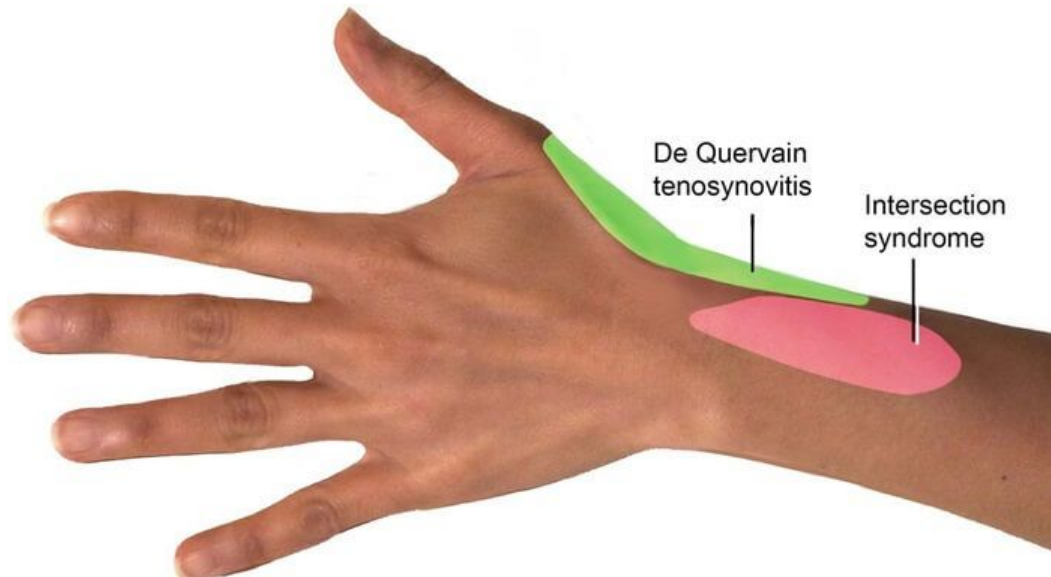
DQT: About



COMMON OVERUSE INJURIES CAUSING RADIAL WRIST/FOREARM SYMPTOMS

	<i>De Quervain</i>	<i>Intersection Syndrome</i>	<i>Wartenberg Syndrome</i>
Pathology	Friction between the APL and EPB in the first dorsal compartment of the wrist	Friction between the APL and EPB as they move over the ECRL/ECRB in the forearm	Compression of the superficial radial nerve between the ECLR and BR in the forearm
Finkelstein test	Positive for pain	Positive for pain	Produces tingling sensation
Tinel test	Negative	Negative	Positive

APL, abductor pollicis longus; ECRB, extensor carpi radialis brevis; ECRL, extensor carpi radialis longus; EPB, extensor pollicis brevis.

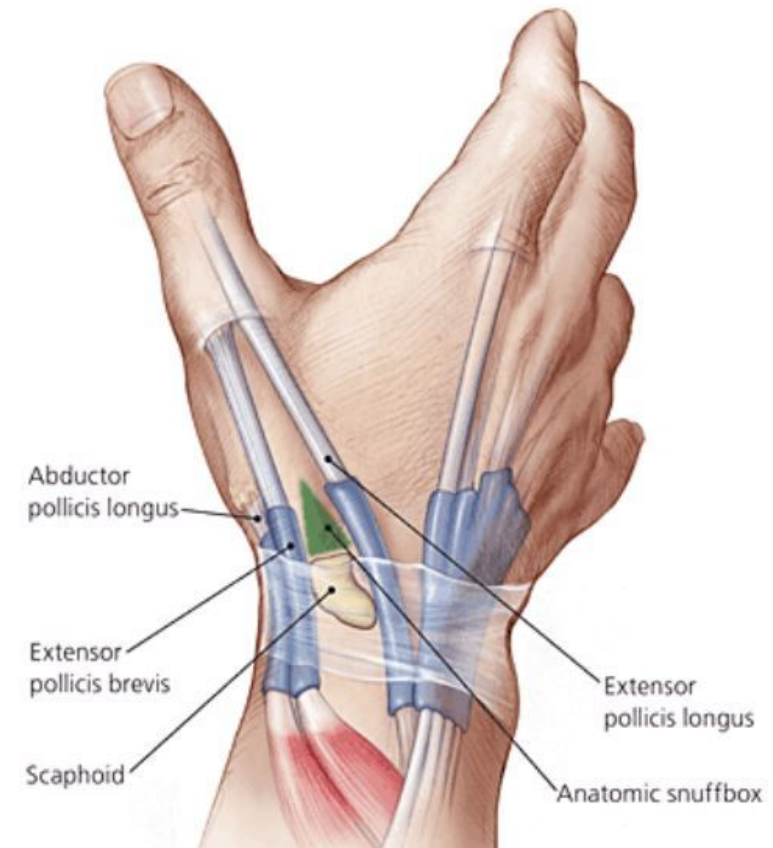


DQT: History

Gradual onset

Overuse of wrist, especially ulnar deviation and thumb movements

Paresthesia is not expected and would suggest Wartenberg syndrome



DQT: Physical assessments

ROM:

- AROM would show pain with ulnar deviation and flexion of the wrist and movements of the thumb
- PROM would be similar to AROM but may not be as provocative
- RROM of APL & EPB would generally be painful but not reproduce symptoms in a neutral position to the same extent as AROM (when the tendons are moving through the irritated sheath).

DQT: Physical assessments

Muscle testing:

- Muscle weakness may occur in abductor pollicis longus and extensor pollicis brevis; MMT of these muscles may rule out muscle strain.

Special tests:

- Finkelstein test will be positive
- Joint play assessment of the wrist may be helpful. If inadequate joint play for deviations is present, motions of deviation will require greater muscle effort and are therefore more likely to produce symptoms. Restoring ulnar glide and radial glide of the wrist may provide important pain relief.

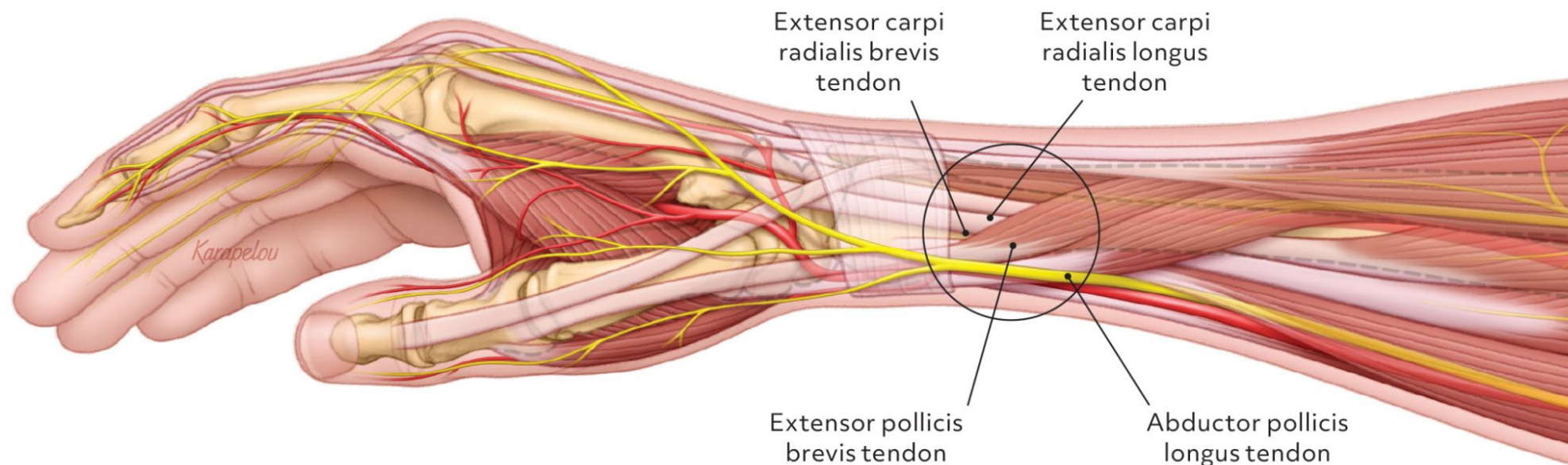
DQT: Management

Identify and modify irritating activity

Orthoses to protect / prevent movement

Strengthening - symptom-free isometrics before isotonic

Avoid compression over lateral radius, especially if paresthesia present



DQT: Massage treatment

Overview of treatment goals for De Quervain's tenosynovitis rehabilitation

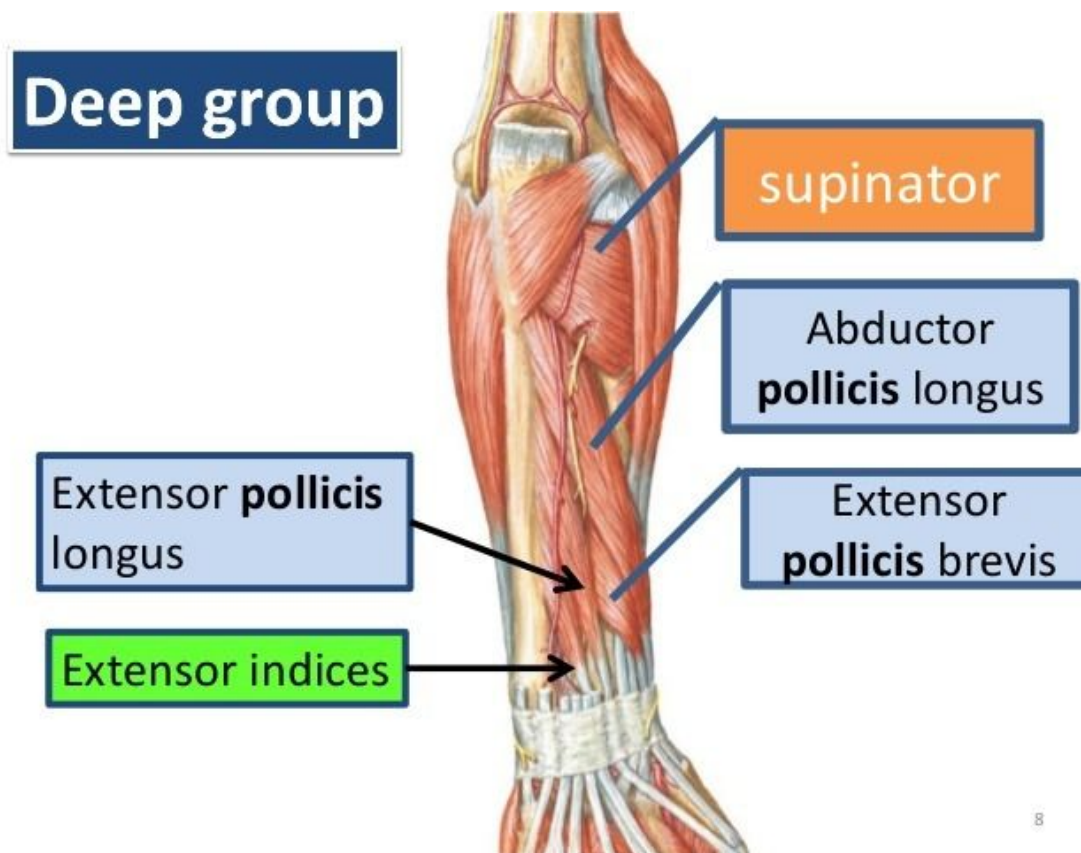
1. Reduce adhesions around the tendon sheath
2. Promote normal resting muscle tone of the affected muscles
3. Promote normal joint play and reduced fascial restrictions in area
4. Treat compensatory structures to restore overall function

Cautions

Do not friction or put direct pressure down into the affected tendon sheath
 Avoid active or passive movement of the thumb unless clearly pain-free; be careful with range of movement
 Avoid heat directly over the tendon sheath which may be inflamed

DQT: Expected outcomes

Provided that the provoking activities are identified and can be reduced or adequately modified, most people have symptoms resolved within 3-6 weeks.



Review:

Turn to your partner.

One person describe Dupuytren contracture - what are some common history and physical assessment findings, how do you approach treatment

One person describe De Quervain's tenosynovitis - what are some common history and physical assessment findings, how do you approach treatment



TFCC Injury

TFCC: About

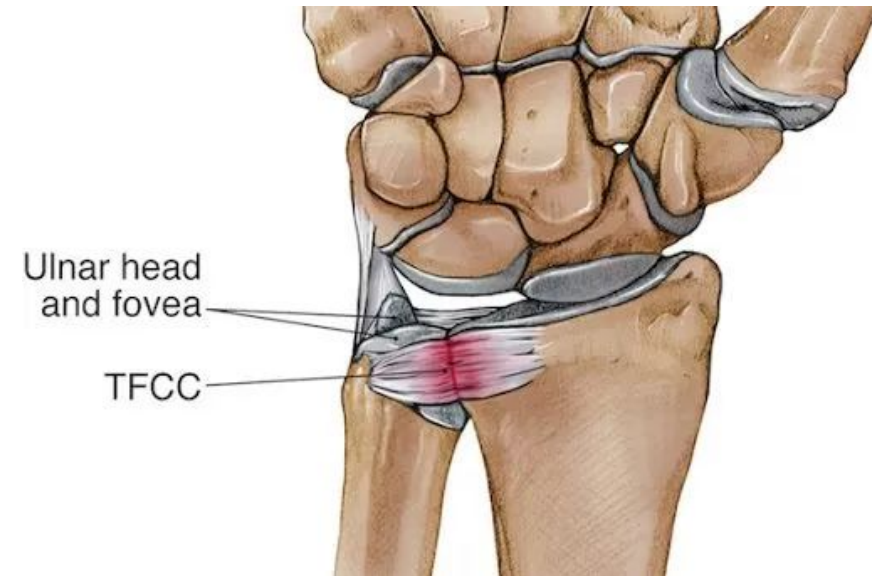
Common cause of ulnar wrist pain

Traumatic

- FOOSH
- Rotation injury

Degenerative

- Repetitive motions (ulnar dev)
- Repetitive motions (extension, pronation)



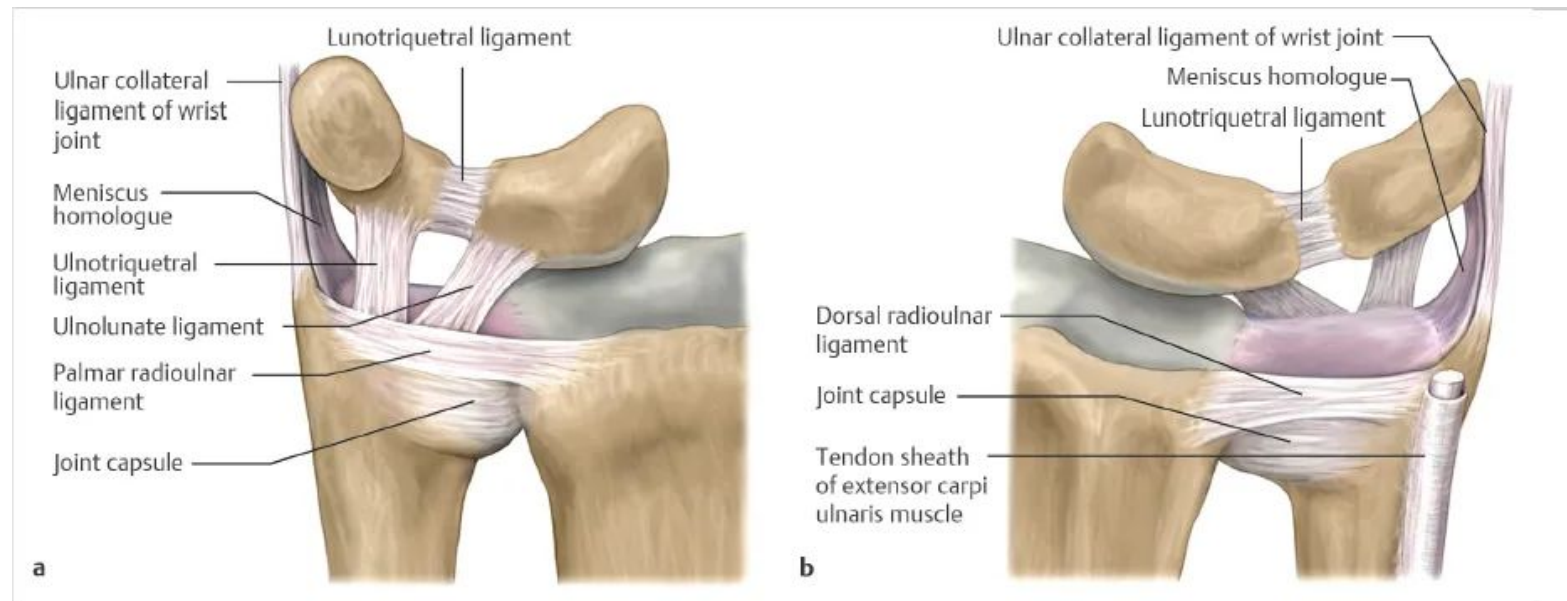
TFCC: History

Deep ulnar sided wrist pain

Worse with weight bearing on hands or carrying heavy objects (strong traction)

Pain or difficulty end range pronation, extension, ulnar deviation

Wrist feels unstable



TFCC: History

The TFCC is poorly vascularised

What implications does this have for patient presentation?

1 /

2 /



TFCC: History

Key questions:

- Did you hear a snap or pop?
- Were you able to continue activity?
- Does the wrist click, lock, or catch?
- Does the wrist feel unstable? Are you able to bear weight through the wrist?
- Did your wrist become swollen and if so, how badly and for how long?

TFCC: Physical assessments

ROM:

- AROM may provoke pain at end-range pronation and ulnar deviation
- PROM is the same as AROM, with overpressure of pronation and ulnar deviation most likely to provoke. End range of radial deviation should not provoke pain in TFCC injury, so if ulnar sided wrist pain occurs with overpressure of radial deviation, it is more suggestive of an ulnar collateral ligament of the wrist sprain as opposed to a TFCC injury
- RROM may show weakness due to pain or inhibition

TFCC: Physical assessments

Muscle testing:

- Muscle testing may be used to rule out injury to FCU or ECU

Special tests:

- Ulnocarpal stress test and ulnar impaction test are positive.



TFCC: Management

Wrist support

- splints / orthoses

ROM and gradual progressive strengthening

NSAIDs and corticosteroids may be used

Surgery possible but normally not attempted until after unsuccessful physical rehab



TFCC: Massage treatment

Overview of treatment goals for TFCC injury rehabilitation

1. Restore neuromuscular control and resting muscle tone of the muscles around the joint
2. Restore normal range of motion of the wrist
3. Treat compensatory structures to restore overall function

Cautions

Avoid joint mobilisation if wrist unstable
 Do not perform axial compression or overpressure in ulnar deviation, extension, pronation
 Avoid joint play of the medial carpals
 Use caution with distraction of the wrist...if carrying heavy objects causes pain, do not attempt traction/distraction

TFCC: Expected outcomes

Prognosis for a TFCC injury is unclear and difficult to provide. Anecdotal experiences range from weeks to over a year of requiring joint support and activity modification.



Fracture

#: About



Common

Distal radius

- Smith's
- Colle's

Scaphoid

Hand (metacarpal)

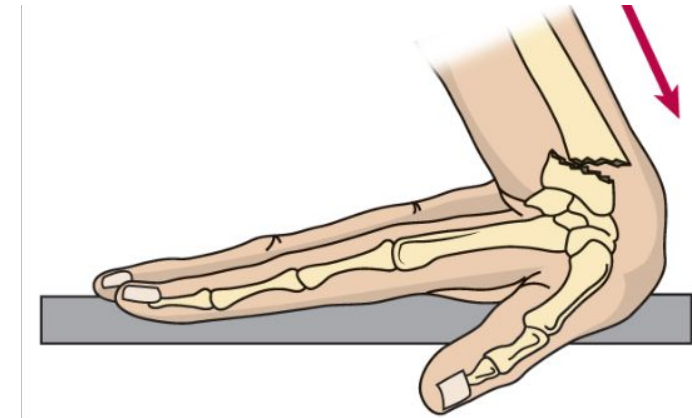


#: About



Colle's

- Distal radius
- Dorsal displacement of distal fracture segment
- Dinner fork appearance
- MOI FOOSH on extended wrist



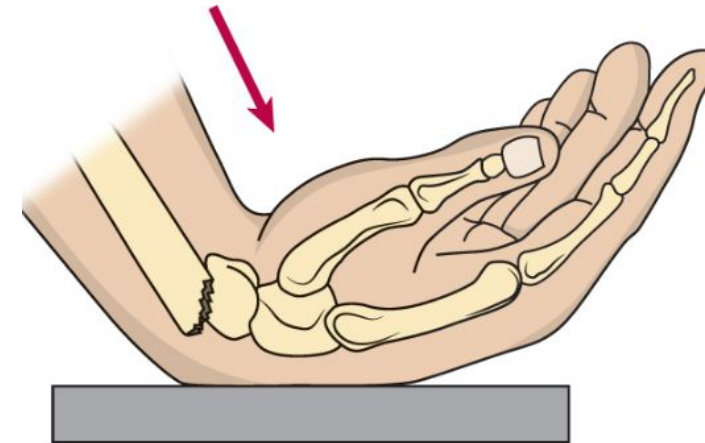
Colles Fracture



#: About

Smith's

- Distal radius
- Volar displacement of distal fracture segment
- Dinner spoon appearance
- MOI FOOSH on flexed wrist



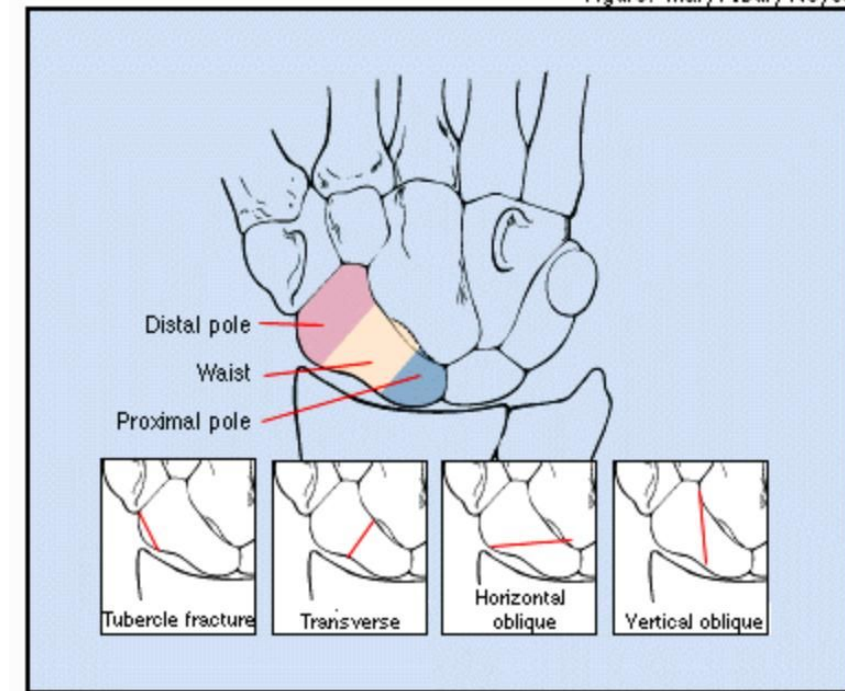
Smith Fracture



#: About

Scaphoid:

- MOI FOOSH
- Possible avascular necrosis (predisposes to arthritis)



#: About

Hand (metacarpal):

- MOI fall, trauma, crush injury
- Swelling may or may not occur
- Generalised hesitation to active motion is a strong sign of hand fracture



#: About

Surgery

- ORIF
- Early movement can begin straight away

Non-surgical

- Cast / immobilisation
- No movement for 4-6 weeks typically

#: History



Following fracture

- Edema
- Limited ROM
- Limited joint mobility
- Muscle weakness
- Secondary problems in shoulder and elbow function

#: History

Key questions:

- Have you had consolidation (full radiographic healing) confirmed by MD?
- Has your doctor cleared you to receive massage?
- Did you have surgery?
- Is there any internal fixation (rods, screws, pins, plates)?
- Is there a surgical scar? How does the scar feel?
- Were you immobilised? In what position? For how long? What type? When was it removed?
- Do you know of any other tissue injury, such as muscle, ligament, blood vessel, or nerve?
- Have you noticed any changes to sensation in the area?
- How is the tissue quality? Does the skin feel healthy? Have you noticed any muscle atrophy?
- Have you received any other treatment? Can you describe it and whether it has helped or not?

#: Physical assessments

ROM (chronic):

- AROM is usually reduced in all ranges, with greatest reductions generally seen in wrist ROM and in supination
- PROM is similar to AROM
- RROM is reduced in all ranges



#: Management

Treat as per general fracture treatment

If surgery was performed, learn about and follow any post-surgical instructions

#: Massage treatment



Overview of treatment goals for post-immobilisation fracture rehabilitation

1. Improve muscle and joint health
2. Restore neuromuscular control and resting muscle tone of area
3. Reduce contractures or adhesions and promote functional scar tissue
4. Treat compensatory structures to restore overall function

#: Massage treatment



Cautions	Pre-consolidation or when consolidation is unknown or unconfirmed: Do not traction the affected bone Do not do POP in either assessment or treatment Do not do deep, aggressive techniques Do not perform joint mobilisation or stretching Pre- or Post-consolidation when working with internal fixation: Modify hydrotherapy to avoid prolonged applications of extreme temperatures Pre- or Post-consolidation when working with low tone tissue, atrophied muscle, or poor tissue health: Avoid deep, longitudinal strokes or aggressive manipulation of tissue Avoid MFR if skin health is poor Do not stretch atrophied muscle
-----------------	--

#: Expected outcomes

Restoration of full ROM may take 4-6 weeks. Restoration of strength - grip strength, intrinsic hand muscles - takes longer and may require up to 12 weeks.

Review:

Turn to your partner.

One person describe TFCC Injury - what are some common history and physical assessment findings, how do you approach treatment

One person describe fracture - what are some common history and physical assessment findings, how do you approach treatment (feel free to look up General Orthopedic treatment notes)

Strains, Sprains, Tendinopathies, Bursitises

Strain, Tendinopathy, Bursitis



Not common

Other than the listed conditions, general strain, tendinopathy, and bursitis at the wrist are not common

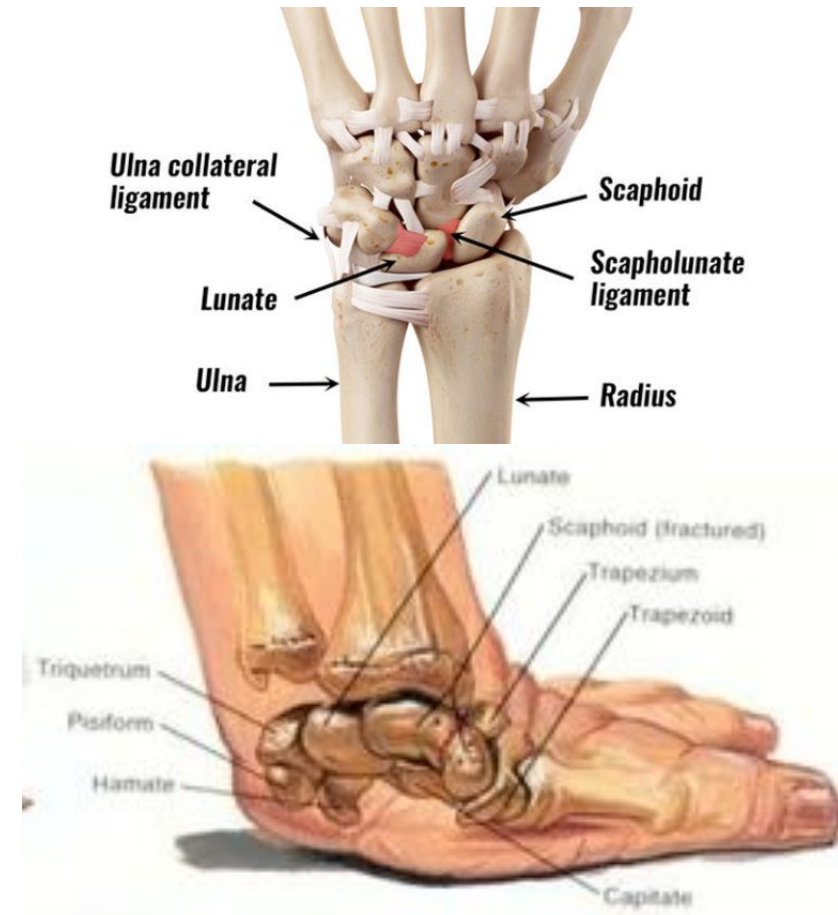
Sprains

FOOSH injuries

Very common and frequently may lead to a palmar ligament sprain (either extrinsic or intrinsic) with or without fracture.

Overpressure in the direction that stresses the suspected ligament (flexion, extension, ulnar deviation, radial deviation) is the best way to assess for the presence of a sprain

Treat as for general sprain





Differential Diagnosis

Radial sided pain

- Carpometacarpal joint arthritis
- Flexor pollicis longus tenosynovitis
- Ulnar collateral ligament sprain
- Carpal tunnel syndrome

Ulnar sided pain

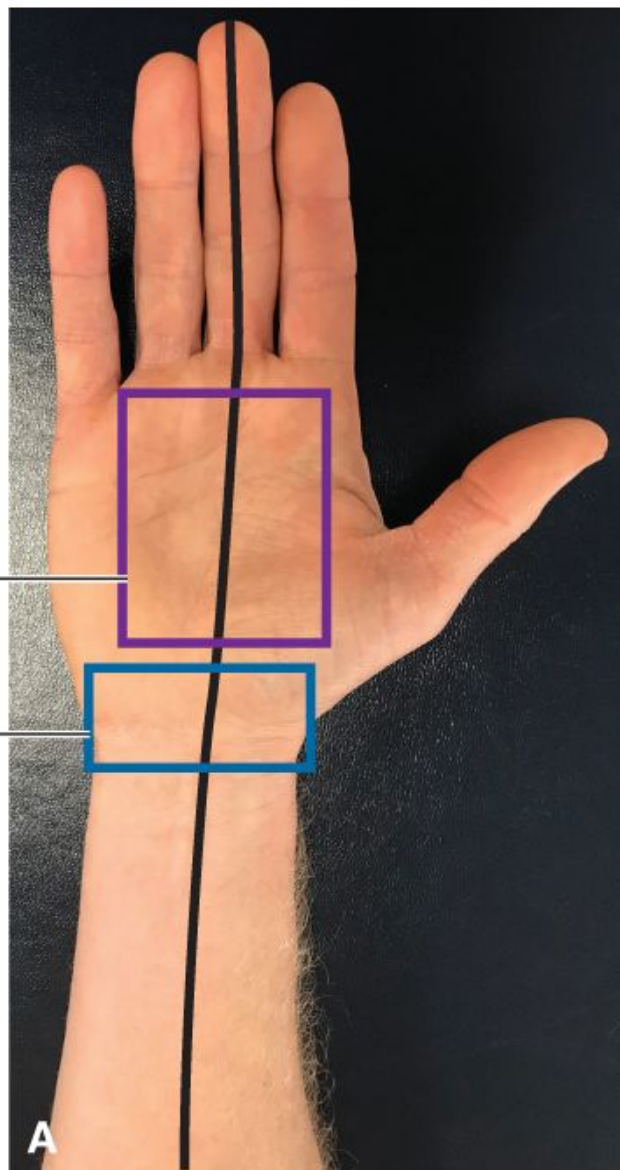
- Triangular fibrocartilage complex injury
- Ulnar nerve compression

Central palm

- Dupuytren contracture
- Trigger finger

Wrist

- Carpal tunnel syndrome
- Wrist instability
- Triangular fibrocartilage complex injury



Radial sided pain

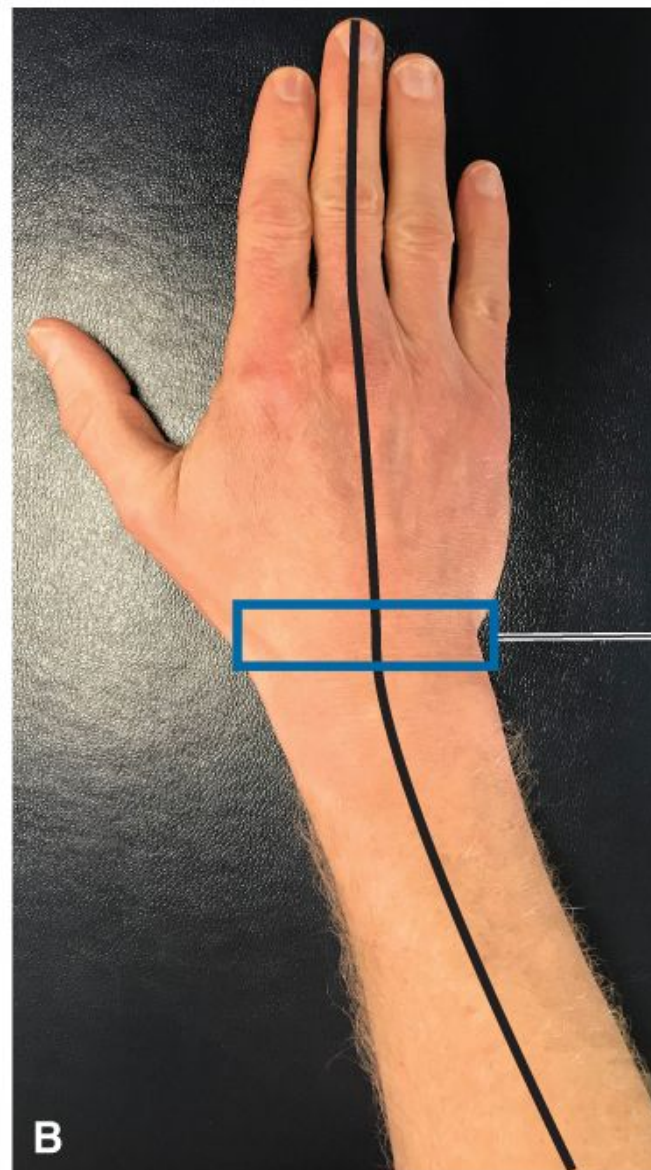
- de Quervain tenosynovitis
- Intersection syndrome
- Wartenberg syndrome
- Ulnar collateral ligament sprain

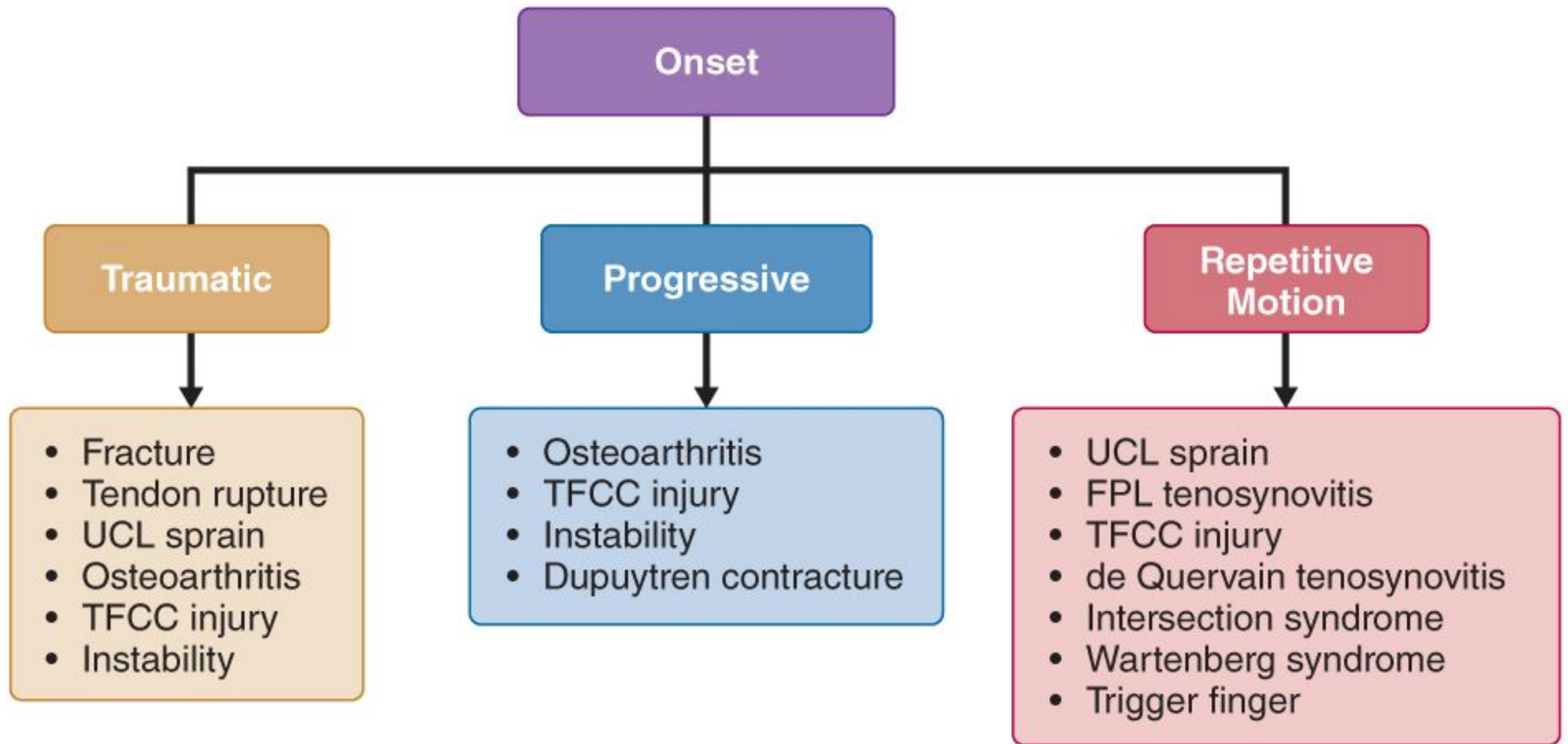
Ulnar sided pain

- Triangular fibrocartilage complex pathology

Wrist

- Wrist instability





DIFFERENTIAL DIAGNOSIS OF WRIST AND HAND PATHOLOGY BY COMPLAINT

<i>Complaint/Symptom</i>	<i>Differential Diagnoses</i>
Weak	Tendon rupture Osteoarthritis Peripheral nerve injury
Paresthesias	Cervical dysfunction Carpal tunnel syndrome Wartenberg syndrome Ulnar nerve compression
Unstable, catching	TFCC pathology Scapholunate ligament sprain Lunotriquetral ligament sprain Midcarpal instability Ulnar collateral ligament sprain
Stiffness	Osteoarthritis Post-immobilization
Difficulty opening fingers or hand	Dupuytren contracture Trigger finger Carpometacarpal joint osteoarthritis Finger osteoarthritis
Deformity	Osteoarthritis Rheumatoid arthritis Extensor tendon disruption Peripheral nerve injury
Pain with gripping	Flexor pollicis longus tenosynovitis Carpometacarpal joint osteoarthritis
Pain with ulnar deviation	de Quervain tenosynovitis Intersection syndrome Wartenberg syndrome TFCC pathology
Pain with weight bearing	TFCC pathology Scapholunate ligament sprain Lunotriquetral ligament sprain Midcarpal instability

TFCC, trianquar fibrocartilage complex.



Examinable conditions

Condition	Written test	OP exam
Wrist sprain	✓	✗
UCL of MCP 1 Sprain	✓	✗
FPL tenosynovitis	✓	✗
CMC 1 Osteoarthritis	✓	✓
Finger arthritis (RA, OA)	✓	✗
Dupuytren's contracture	✓	✓
Trigger finger	✓	✗
De Quervain's tenosynovitis	✓	✓
TFCC injury	✓	✗
Fracture	✓	✗

Learning Objectives. Can you...?



- Describe all special tests (indication, procedure, positive finding, explanation of results) in special tests document
- Perform all special tests in special tests document using full NEER protocol
- Describe definition, MOI, affected tissue, signs and symptoms for UCL of MCP 1 sprain; FPL tenosynovitis; CMC 1 Osteoarthritis; Finger arthritis (OA & RA); Dupuytren contracture; Trigger finger; De Quervain's tenosynovitis; TFCC injury; Fracture
- Describe treatment plan (history, physical assessments, clinical impression and approach to treatment, precautions, and home care) for UCL of MCP 1 sprain; FPL tenosynovitis; CMC 1 Osteoarthritis; Finger arthritis (OA & RA); Dupuytren contracture; Trigger finger; De Quervain's tenosynovitis; TFCC injury; Fracture